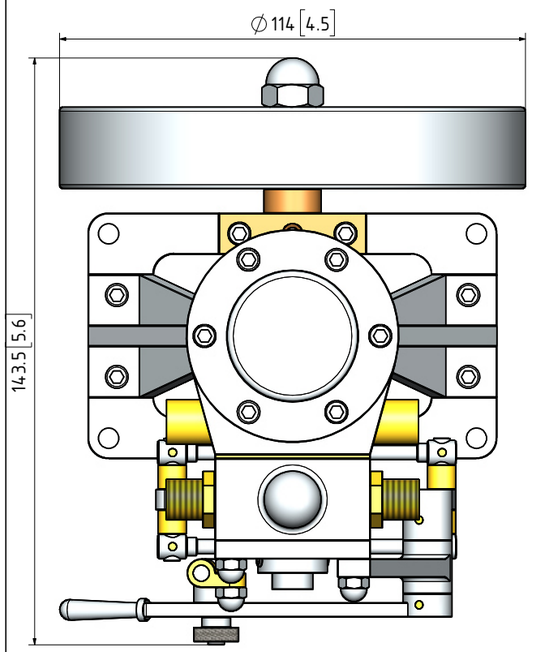
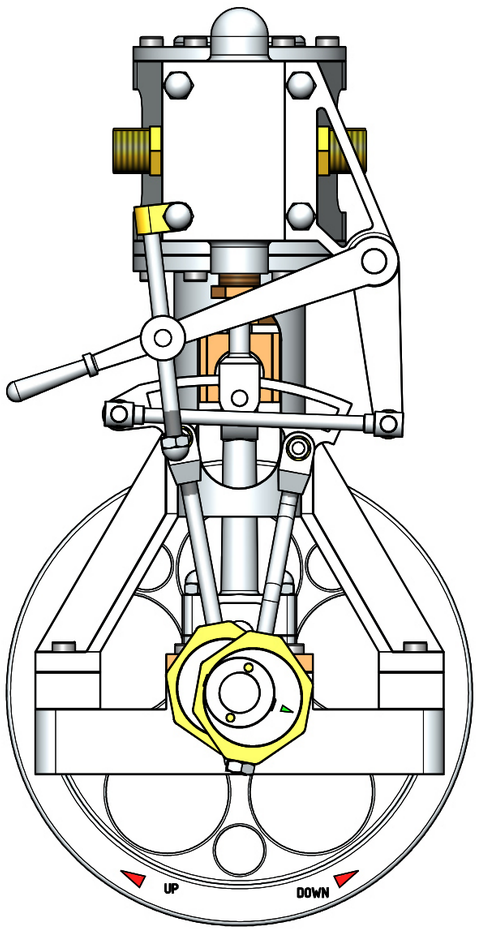
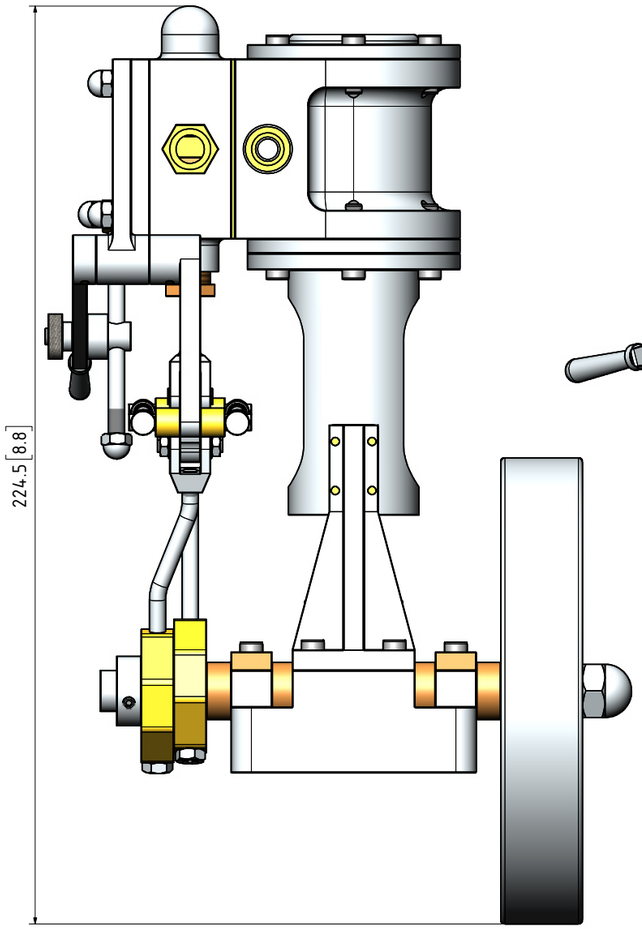


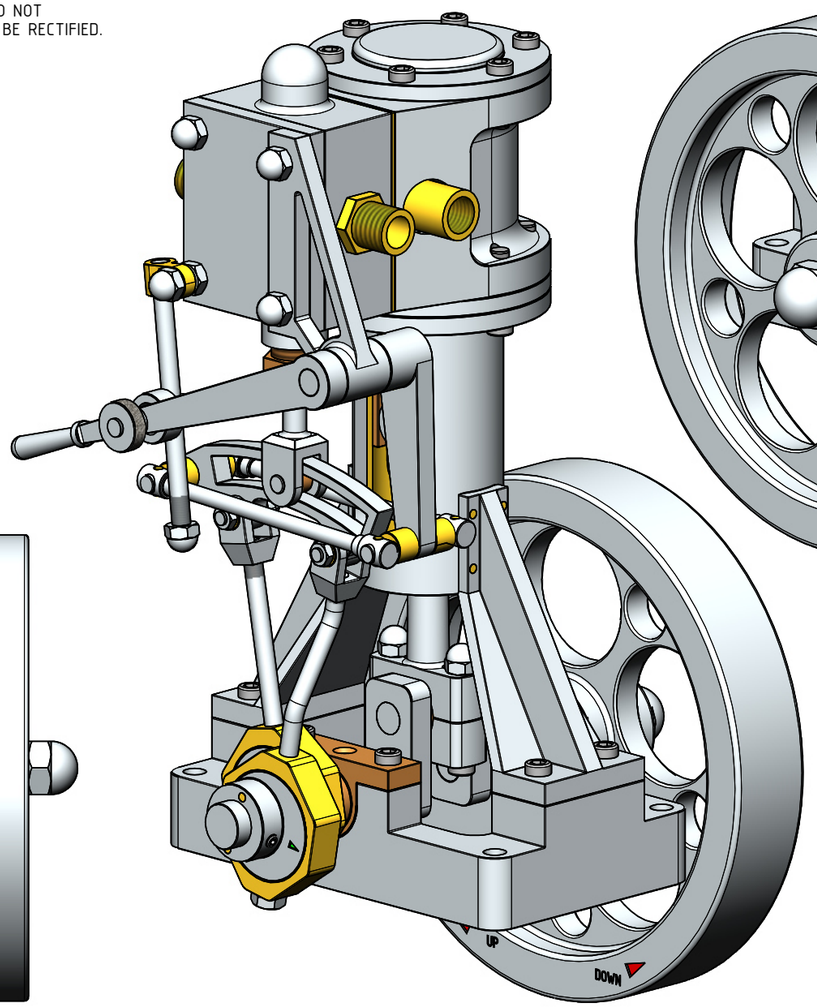

NOTES:
0. ALL DRAWINGS ARE IN METRIC MEASUREMENTS
1. ALL ENGINEERING PRACTICES SHALL BE APPLIED WITH REGARDS TO HOLE AND SHAFT TOLERANCES.
2. WHERE SCREWS OR BOLTS ARE USED THE CLEARANCE HOLES SHALL BE APPROXIMATELY 5% TO 8% LARGER THAN THE MATCHING TAPPED HOLE.
3. PREFERABLY ALL TAPPED HOLES AND MATCHING SCREWS AND/OR BOLTS TO BE METRIC FINE (MF)
4. MATERIALS SPECIFIED ON THE DRAWINGS ARE INDICATIVE ONLY. THE BUILDER CAN MAKE HIS/HER OWN MATERIAL CHOICE.
5. ALL CONNECTIONS/JOINTS WHICH HAVE STEAM PRESSURE APPLIED TO IT SHALL BE SILVER/HARD SOLDERED.
6. COMPRESSION SPRINGS ARE DRAWN IN COMPRESSED STATE (CP), UNCOMPRESSED STATE IS APPROX 40% TO 60% LONGER THEN COMPRESSED STATE.
7. WHERE PREFERRED SCREW OR RIVETED CONNECTIONS CAN BE OMITTED AND PARTS CAN BE BONDED TOGETHER BY USING EITHER HIGH STRENGTH GLUE, EPOXY RESIN, OR SOLDER.
8. PARTS WHICH ARE DIRECTLY EXPOSED TO STEAM AND/OR WATER SHOULD BE CONSTRUCTED USING NON-FERROUS OR NON CORROSIVE MATERIAL SUCH AS BRASS, BRONZE, GUNMETAL, STAINLESS STEEL, COPPER OR MONEL.
9. THE ORDER IN WHICH THE PARTS/COMPONENTS ARE MANUFACTURED AND THE MODEL IS ASSEMBLED IS ENTIRELY LEFT TO THE BUILDER/MODEL MAKER.
10. A COLOUR SCHEME FOR THIS PROJECT IS ENTIRELY LEFT UP TO THE MODEL MAKER.
11. THE MANNER IN WHICH THE PARTS/COMPONENTS ARE MANUFACTURED IS ENTIRELY LEFT UP TO THE BUILDER.
12. USE LOCTITE, ON SCREW OR PRESS FIT CONNECTIONS OR SURFACES, WERE DEEMED NECESSARY TO PREVENT PARTS FROM LOOSENING.
13. WASHERS AND/OR SPRINGWASHERS SHALL BE USED WHERE DEEMED NECESSARY.
XX. ERRORS AND/OR OMISSIONS MAY OCCUR IN THE DRAWINGS, DO NOT HESITATE TO CONTACT ME SO THAT THE ERRORS/OMISSIONS CAN BE RECTIFIED.

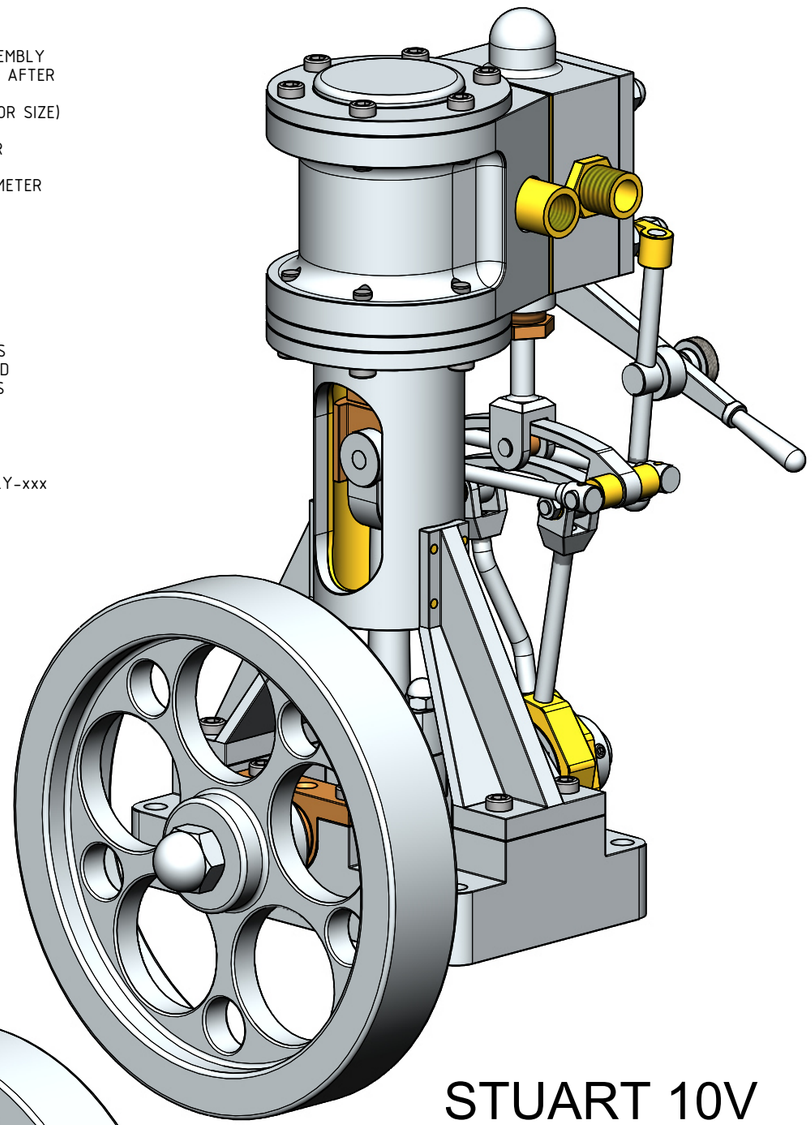
MATERIAL ABBREVIATIONS:
ALU = ALUMINIUM
HALU= HARD ALUMINIUM
BRS = BRASS
BRZ = BRONZE OR GUNMETAL (BRZ/GM)
CI = CAST IRON
CU = COPPER
GRA = GRAPHITE
MS = MILD STEEL/BRIGHT MILD STEEL
SS = SILVER STEEL OR STAINLESS STEEL
SPS = SPRING STEEL
PEEK= POLYETHER ETHER KETONE
SYN = SYNTHETIC MATERIAL SUCH AS VETON, NYLON, TEFLON OR RUBBER
IN GENERAL SYNTHETIC MATERIALS SOULD BE ABLE TO WITHSTAND THE HEAT AND PRESSURE(S) APPLIED TO THEM.
nnn/nnn MEANS THAT EITHER MATERIAL CAN BE USED

OTHER ABBREVIATIONS
AS = AS SHOWN
DP = DEEP
DAA= DRILL AFTER ASSEMBLY
D&TAA= DRILL AND TAP AFTER ASSEMBLY
CF = CLOSE FIT (SIZE FOR SIZE)
PF = PRESS FIT
PFAA= PRESS FIT AFTER ASSEMBLY
PCD = PITCH CIRCLE DIAMETER
RM = REAM
HEX = HEXACON, 6SIDED
CP = COMPRESSED
KNL = KNURLED
CSK = COUNTERSINK
PL = PLACES
DWL= DOWEL
SPF= SPOTFACE
(T)HESOP=(TAPPED)HOLES EQUALLY SPACED ON PCD
(T)HESOC=(TAPPED)HOLES EQUALLY SPACED ON CIRCUMFERENCE
OD = OUTSIDE DIAMETER
ID = INSIDE DIAMETER
[SA-xxx]= SUB ASSEMBLY-xxx



# STUART 10V

| QTY. | PART NUMBER |
|---|---|
| 1 | 09C-50-00-1-01-BASE BLOCK |
| 1 | 09C-50-00-1-02-CYLINDER BLOCK |
| 1 | 09C-50-00-1-03-CYLINDER BOTTOM COVER |
| 1 | 09C-50-00-1-04-CYLINDER TOP COVER |
| 1 | 09C-50-00-2-01-CRANK SHAFT + FLYWHEEL |
| 1 | 09C-50-00-2-02-PISTON+CROSSHEAD |
| 1 | 09C-50-00-2-03-CON-ROD |
| 1 | 09C-50-00-2-04-SLIDE VALVE+ROD |
| 1 | 09C-50-00-2-05-ECCENTRIC SHEAVES |
| 1 | 09C-50-00-2-06-EXPANSION LINK |
| 1 | 09C-50-00-2-07-RESTRICTER BAR |
| 1 | 09C-50-00-2-08-RESTICTER CLAMP |
| 1 | 09C-50-00-2-09-REVERSING HANDLE |
| 1 | 09C-50-00-2-10-LINK RODS |
| 1 | 09C-50-00-2-11-ECCENTRIC STRAP |
| 1 | 09C-50-00-2-12-OUTER ECCENTRIC STRAP |
| 4 | 09C-50-00-M3 NUT |
| 4 | 09C-50-00-M3 WASHER |
| 2 | 09C-50-00-M3x4 A-K GRUB SCREW |
| 7 | 09C-50-00-M4 DOME NUT |
| 3 | 09C-50-00-M4 NUT |
| 14 | 09C-50-00-M4x12 A-K CYL HEAD SCREW |
| 6 | 09C-50-00-M4x16 A-K CYL HEAD SCREW |
| 2 | 09C-50-00-M4x24 A-K CYL HEAD SCREW |
| 1 | 09C-50-00-M8 DOME NUT |

NOTES: THIS DESIGNE IS LOOSELY BASED ON DRAWINGS WHICH WERE GIVEN TO ME. THE DESIGNER OR PUBLISHING MAGAZINE OF THE ORIGINAL DRAWINGS IS UNKNOWN. THE ONLY REFERENCE IS A DRAWINGNUMBER: DRG. 70207. THIS ENGINE IS 1.5 TIMES LARGER THAN THE ORIGINAL

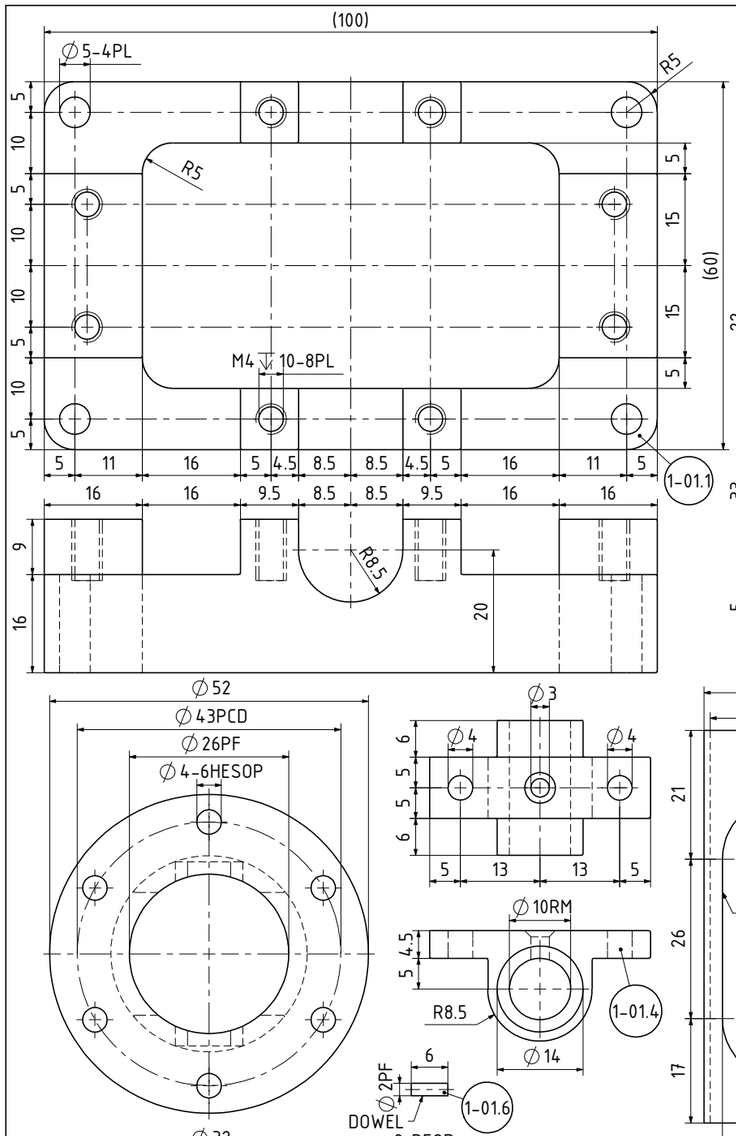
TITLE
1 CYLINDER VERTICAL STEAM ENGINE WITH REVERSER (BORE=30mm x STROKE=28mm)

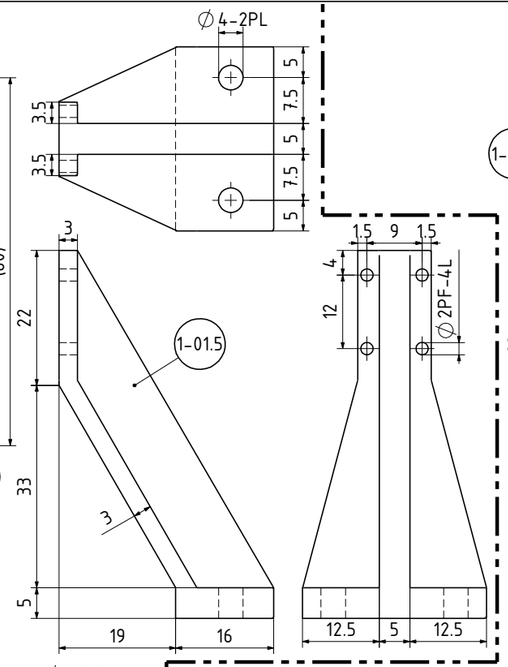
DRAWING CONTENTS
GENERAL ARRANGEMENT, ISOMETRIC VIEWS, NOTES, BILL OF MATERIALS

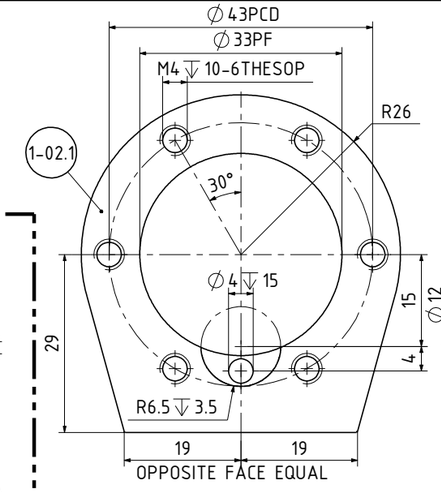
PROJECT No 09C-50-00
JDW DRAUGHTING SERVICES
J.A.M. DE WAAL. 12 BRIGHTWELL STREET PAPAKURA 2110. NEW ZEALAND. PHONE: 0064 09 2988815. MOB: 0211791000 E-MAIL: dewaal@xtra.co.nz.

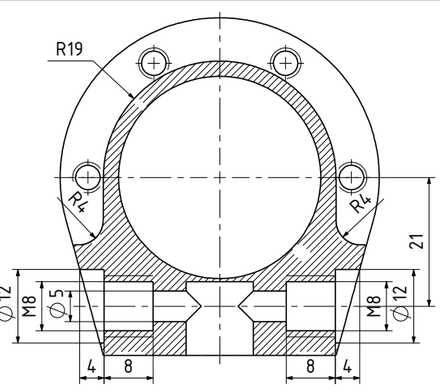
PROJECTION
JDWDS
MODEL SCALE: 1:1
DWG SCALE: 1:1 @A3 OR AS SHOWN
DATE SEPTEMBER 2020
Copyright © J.A.M. DE WAAL PAPAKURA NZ
SHEET: 01 OF 04
A3
No: 09C-50-00-SHT01

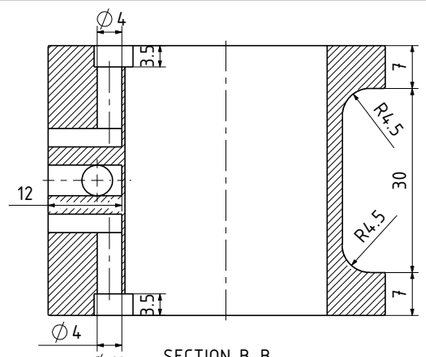
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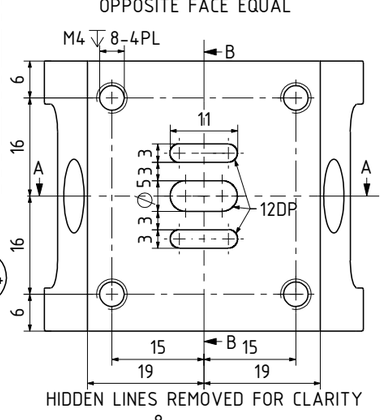


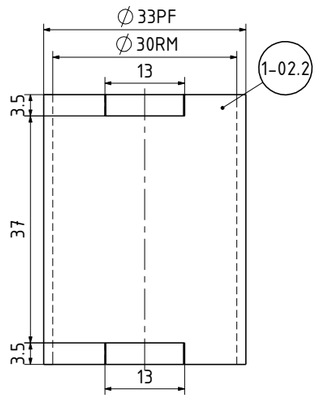



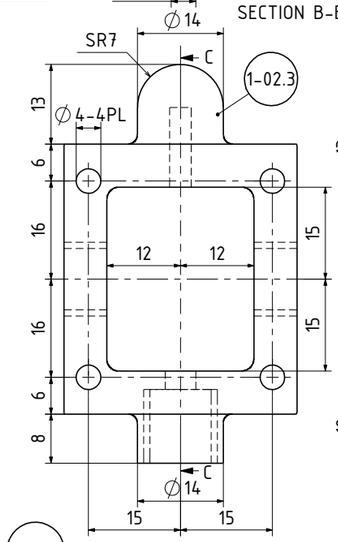



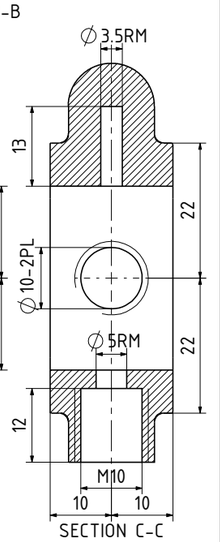



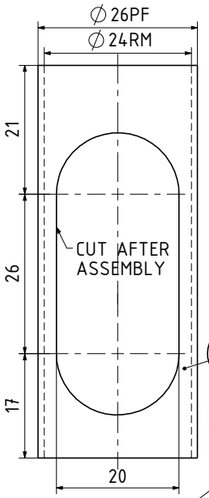



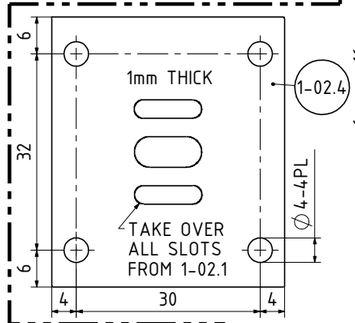



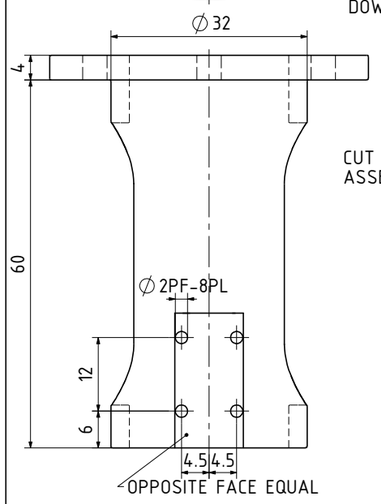


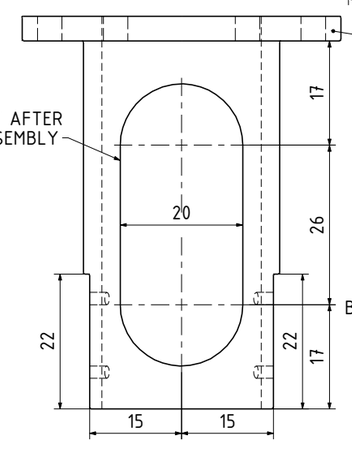


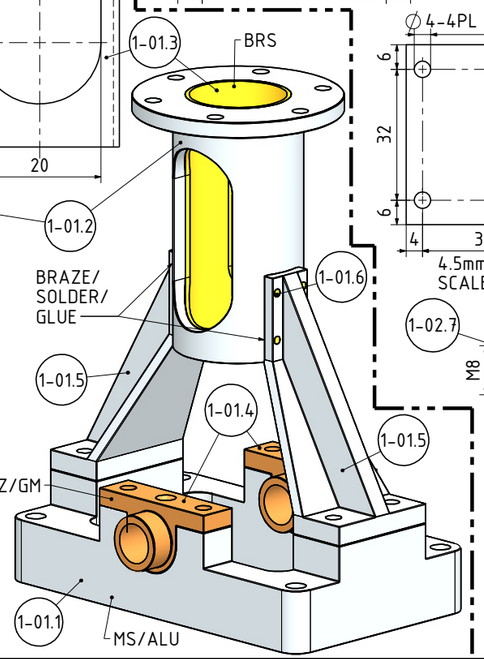


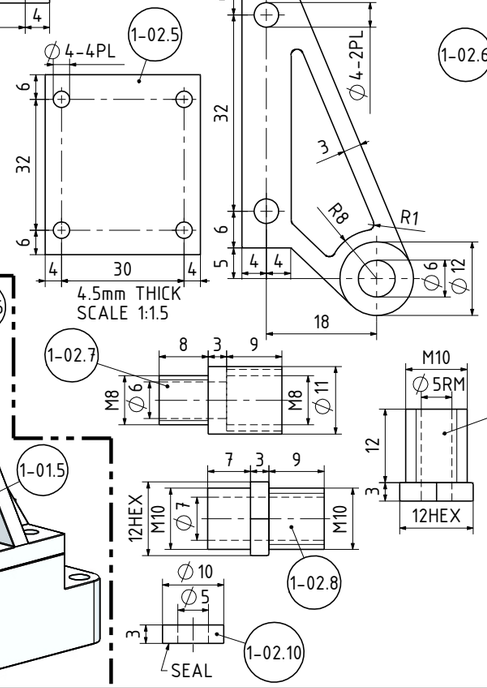



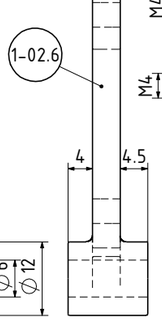



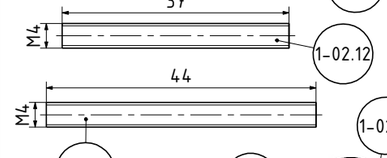



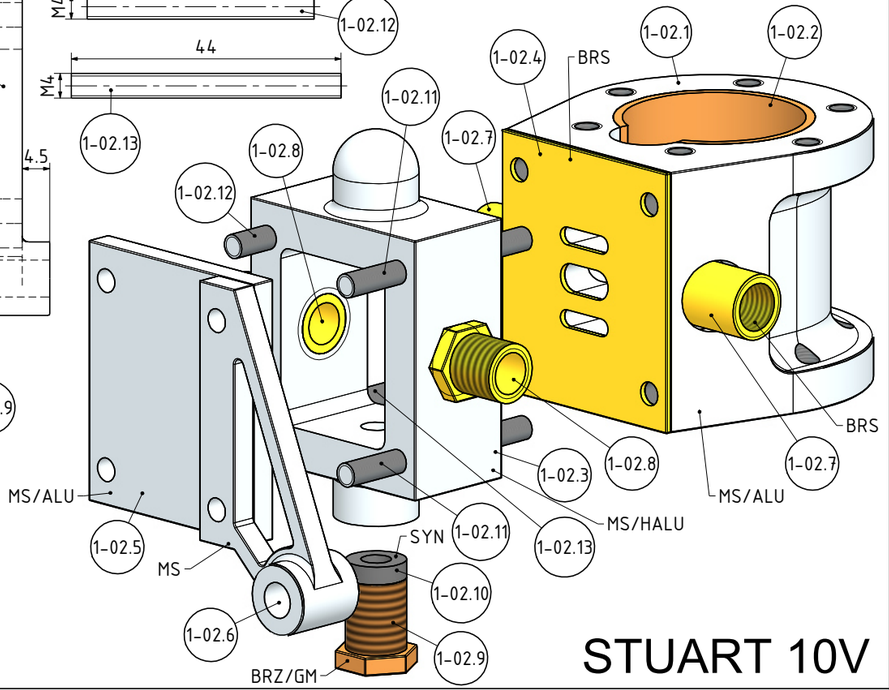




# STUART 10V

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TITLE

## 1 CYLINDER VERTICAL STEAM ENGINE WITH REVERSER (BORE=30mm x STROKE=28mm)

DRAWING CONTENTS

## PARTS AND ASSEMBLIES

PROJECT No 09C-50-00

JDW DRAUGHTING SERVICES
J.A.M. DE WAAL. 12 BRIGHTWELL STREET PAPAKURA 2110. NEW ZEALAND. PHONE: 0064 09 2988815. MOB: 0211791000 E-MAIL: dewaal@xtra.co.nz.

PROJECTION | JDWDS | MODEL SCALE: 1:1

DWG SCALE: 1:1 @A3 OR AS SHOWN

DATE SEPTEMBER 2020

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SHEET: 02 OF 04 | A3 | No: 09C-50-00-SHT02

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SA-1-101

SA-1-105

DETAIL B
SCALE 2 : 1

SECTION A-A
SCALE 1 : 1.5

5mm THICK

5mm THICK

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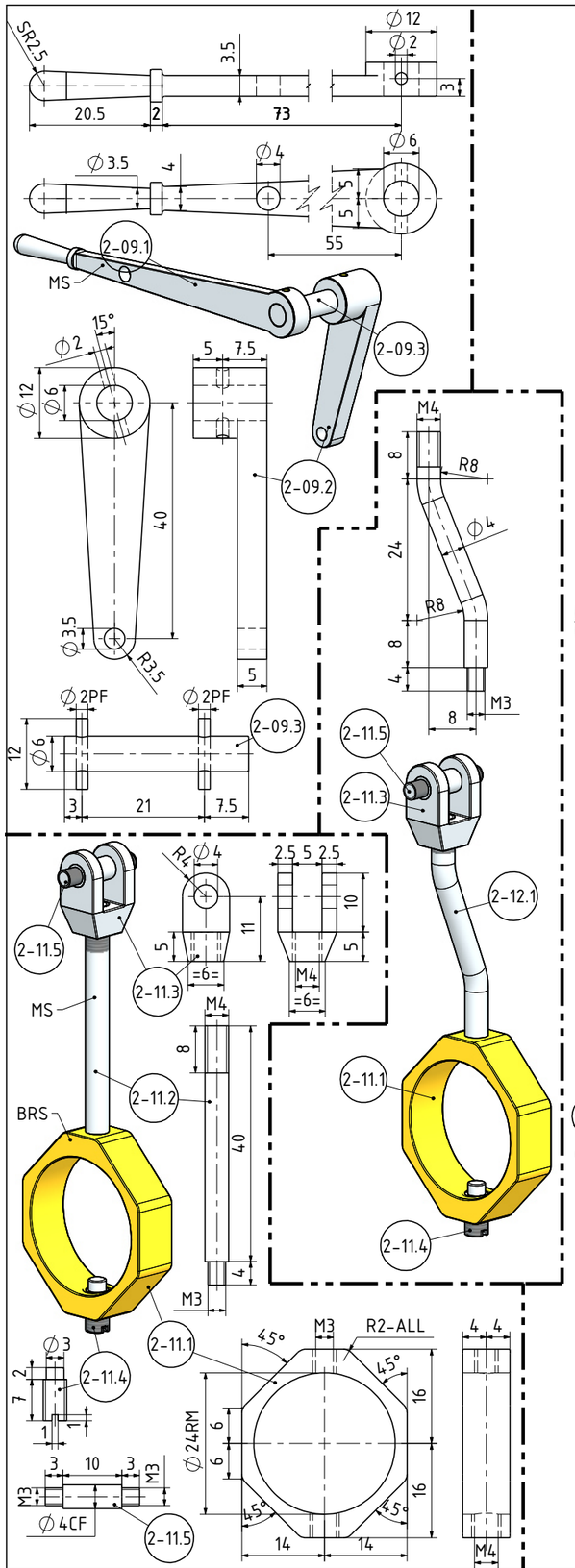
TITLE
# 1 CYLINDER VERTICAL STEAM ENGINE WITH REVERSER (BORE=30mm x STROKE=28mm)

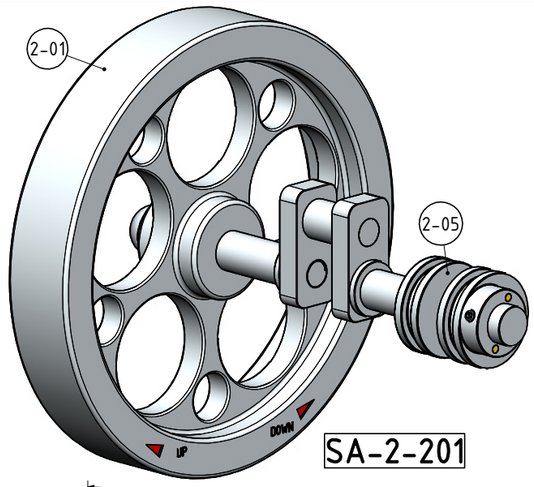
DRAWING CONTENTS
## PARTS AND ASSEMBLIES
## STUART 10V

PROJECT No 09C-50-00
JDW DRAUGHTING SERVICES
J.A.M. DE WAAL. 12 BRIGHTWELL STREET PAPAKURA
2110. NEW ZEALAND. PHONE: 0064 09 2988815. MOB: 0211791000 E-MAIL: dewaal@xtra.co.nz.

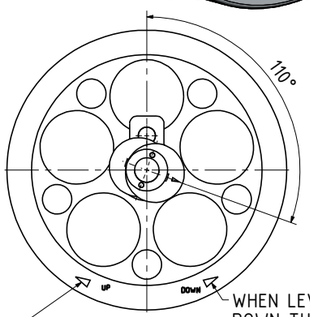
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MODEL SCALE: 1:1
DWG SCALE: 1:1 @A3 OR AS SHOWN
DATE SEPTEMBER 2020
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SHEET: 03 OF 04 A3 No: 09C-50-00-SHT03

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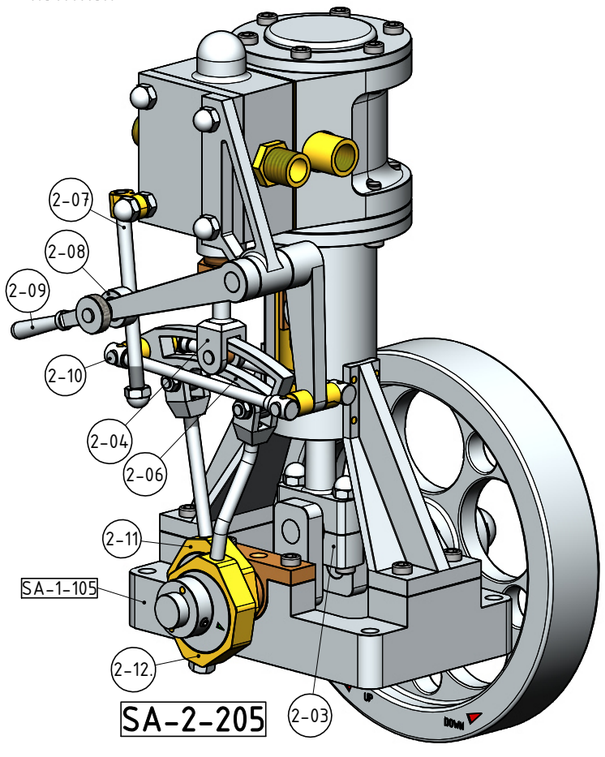




SA-2-201

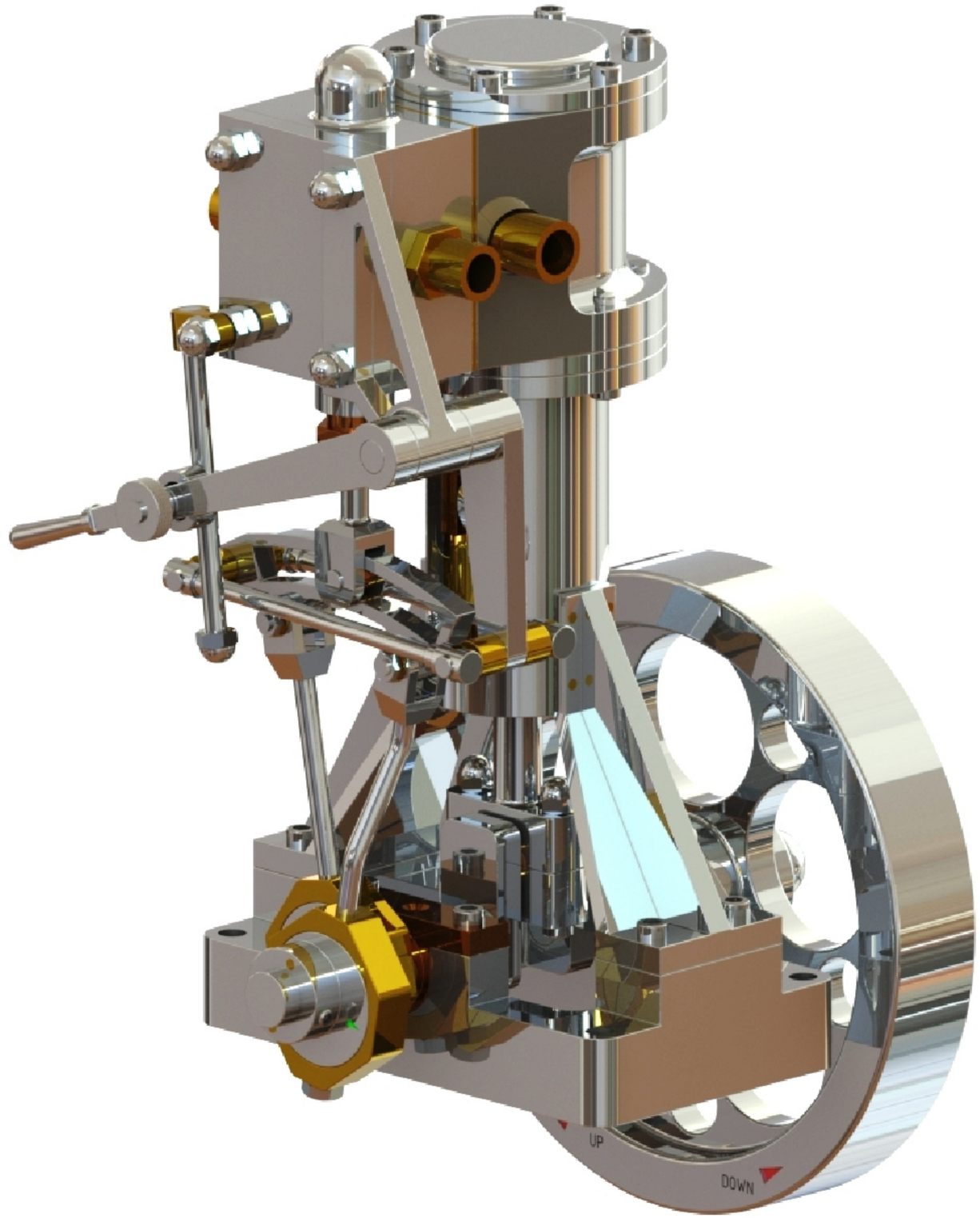



WHEN LEVER IS UP THEN THIS IS DIRECTION OF ROTATION

WHEN LEVER IS DOWN THEN THIS IS DIRECTION OF ROTATION




SA-2-205




# STUART 10V

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TITLE

1 CYLINDER VERTICAL STEAM ENGINE WITH REVERSER (BORE=30mm x STROKE=28mm)

DRAWING CONTENTS

PARTS AND ASSEMBLIES, RENDERED PICTURE


PROJECT No 09C-50-00

JDW DRAUGHTING SERVICES

J.A.M. DE WAAL. 12 BRIGHTWELL STREET PAPAKURA 2110. NEW ZEALAND. PHONE: 0064 09 2988815. MOB: 0211791000 E-MAIL: dewaal@xtra.co.nz.


PROJECTION

JDWDS

MODEL SCALE: 1:1

DWG SCALE: 1:1 @A3 OR AS SHOWN

DATE SEPTEMBER 2020

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SHEET: 04 OF 04

A3

No: 09C-50-00-SHT04

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